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Google scholar –https://scholar.google.com/citations?view_op=list_works&hl=en&user=CSpUmgYAAAAJ
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Area of Interest

- GNSS-Reflectometry, Geodesy, SLR, DORIS
- AI, Data Science, Machine Learning and Deep Learning
- Remote sensing and GNSS Applications

Skill Set

- Skills: GNSS, DORIS, SLR signal processing, Programming Statistics and Probability, Machine Learning, Deep Learning, Data Mining, Data Visualization, Data Manipulation.
- Tools: MATLAB, Python NumPy Pandas SQL Power BI, Tableau, Matplotlib, Seaborn, Plotly, Excel, ChatGPT AI Tools

Awards and Achievements

- Developed **Software** for preprocessing of **NavIC** signal based on **MATLAB** that has been utilized inside the SAC ISRO.
- **GATE qualified** with **99.7** percentile.
- **Innovator of the year award,** 16th Uttarakhand State Science and Technology Congress 2022 at **UCOST Dehradun**.
- **Best paper award** at National seminar organized jointly by **IIRS** (Indian Institute of Remote Sensing) and **ISRO** (Indian Space Research Organization) in 2021.
- **Young Scientist award,** 14th Uttarakhand State Science and Technology Congress 2020 at **UCOST Dehradun**.
- **Best researcher award for Environmental studies** during 3rd world conference on Innovations in Management Science and Engineering.

Education

- PhD from Graphic Era Deemed to be University.
- M.Tech (Microelectronics) from JIIT Noida 62.
- B.Tech (Electronics) Bharati Vidyapeeth College of Engineering, Pune.
- 12th from Holy Mission , Samastipur, affiliated to C.B.S.E.
- 10th from Kendriya Vidyalaya, Barauni, affiliated to C.B.S.E.

Publications

- **SCI** Indexed Journal: **7**
- **Scopus** Indexed Journal: **2**
- **Scopus** Indexed conference: **36**

Experience

- Currently working as **Research Establishment Officer (REO), Grade 1, Level 11** at **National Centre of Geodesy (NCG)**, IIT Kanpur.
- Part time Working as **Python, ML, NLP, Power BI trainer** at **Embedded Technologies** Ghaziabad.
- **2 months** experience as Data Science Intern at **Unmessenger.**
- Two-year experience as **Senior Research Fellow (SRF)** in **ISRO** sponsored project at **Graphic EraDeemed to be University.**
- One year and 10 months experience as **Junior Research Fellow (JRF)** in **ISRO** sponsored project at **Graphic EraDeemed to be University**.
- Two year and 6 months experience as **Assistant Professor** at **Tula's Institute Dehradun, the engineering and Management College**, Approved by AICTE & affiliated to Uttarakhand Technical University.
- Worked as **Teaching assistant** at **JIIT NOIDA** during MTech.
- 6 month of teaching experience as an **Embedded and VLSI Trainer** at **Embedded Technologies** Ghaziabad.

Professional Courses

- **6 months** post graduate certification course on **Artificial Intelligence and Deep Learning** conducted by **IIT Roorkee.**
- Currently perusing **1-year** online certification course on data science **by Dataisgood in association with IBM and Microsoft** along with different courses by Udemy on Machine learning and Data Science.
- Completed online course **"Remote Sensing & GIS Technology and Applications for University Teachers & Government Officials"** which was conducted by **Indian Institute of Remote Sensing (IIRS), ISRO Dehradun**.
- Completed online course of 3-month duration **"Basics of Remote Sensing Geographical Information System and Global Navigation Satellite System"** which was conducted by **Indian Institute of Remote Sensing (IIRS), ISRO Dehradun.**
- Completed online course **"Remote Sensing in Crop Monitoring and Assessment"** which was conducted by **Indian Institute of Remote Sensing (IIRS), ISRO Dehradun.**
- Completed the online course on **"Overview of Planetary Geosciences with special emphasis to the Moon and Mars"** conducted by **Indian Institute of Remote Sensing (IIRS), ISRO Dehradun.**

Disclaimer

I hereby declare that the information above furnished is true to my best knowledge and belief.

PLACE: IIT KANPUR

DATE: 25-01-2026 **SUSHANT SHEKHAR**

List of Key Publications with Peer Review Process

A. Journals

1. Gandugade, P., **Shekhar, S**., Nath, S., Nagarajan, B., & Dikshit, O., 2025. The role of satellite laser ranging in space geodesy: Developments, applications, and the Indian perspective, Current Science, In press.
2. **S. Shekhar**, R. Prakash, D. K. Pandey, A. Vidyarthi, D. Putrevu and N. Desai et al., Multipath phase-based vegetation correction scheme for improved field-scale soil moisture retrieval over agricultural cropland using GNSS-IR technique, Advances in Space Research, 2024. **(SCI)**
3. **S. Shekhar**, R. Prakash, D. K. Pandey, A. Vidyarthi, D. Putrevu and N. Desai et al., (2023) "Wavelet analysis for quantifying NavIC-IR multipath signal frequency with the aim of determining soil moisture in agricultural fields", Remote Sensing Letters, 14:12, 1315-1323. **(SCI)**
4. **S. Shekhar,** R. Prakash, D. Pandey, A. Vidyarthi, D. Putrevu, and N. M. Desai, "Comparative Analysis of NavIC Multipath Observables for Soil Moisture Over Different Field Conditions," Progress In Electromagnetics Research M, Vol. 117, 25-35, 2023. **(SCI)**
5. **S. Shekhar**, R. Prakash, D. K. Pandey, A. Vidyarthi, S. Tyagi, D. Putrevu and A. Misra et al., "Development of Soil Moisture Inversion Model for Bare Soil Using Navigation with Indian Constellation (NavIC)," in IEEE Geoscience and Remote Sensing Letters, vol. 19, pp. 1-5, 2022, Art no. 3003605. **(SCI)**
6. A. Gautam, G.Verma, and **S. Shekhar** "Vehicle Pollution Monitoring, Control and Challan System Using MQ2 Sensor Based on Internet of Things", Wireless Personal Communication 116, 1071–1085 (2021). **(SCI)**
7. G.Verma, T. Singhal, and **S. Shekhar** "Heuristic and Statistical Power Estimation Model for FPGA Based Wireless Systems", Wireless Personal Communication 106, 2087–2098 (2019). **(SCI)**

B. Conferences (IEEE and Scopus indexed which all are peer reviewed)

1. A. Tripathi, S. Shekhar, D. K. Pandey, R. Prakash, A. Vidyarthi, and O. Dikshit, "Improving EOS-04 (RISAT-1A) SAR based Operational 500 m Soil Moisture Product Using Meteorological Observables and Machine Learning Model," in *Proc. IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*, 2024, pp. 1–4.
2. P. Gandugade, S. Shekhar, S. Nath, B. Nagarajan, and O. Dikshit, "Advancing India's Geospatial Science: The Imperative for Satellite Laser Ranging (SLR) Infrastructure," in *Proc. IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*, 2024, pp. 1–4.
3. A. Raj, S. Shekhar, D. K. Pandey, R. Prakash, A. Vidyarthi, and O. Dikshit, "Integrating EOS-04 SAR, SMAP Radiometer and CYGNSS GNSS-R Data with Machine Learning for Improved Soil Moisture Retrieval Accuracy," in *Proc. IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*, 2024, pp. 1–4.
4. Nath S., Prasad, A. V., Shekhar, S., and Dikshit, O., Crustal deformation analysis associated with 2019 Kashmir earthquake using DInSAR technique, 2025 IEEE India Geoscience and Remote Sensing Symposium (InGARSS 2025), 2025.
5. Nath S., Ketholia, Y., Majumder, M., and Shekhar, S., Integrating spaceborne observations for deformation analysis in the Central Himalayas: A case study of 2022 Nepal earthquake, 2025 IEEE India Geoscience and Remote Sensing Symposium (InGARSS 2025), 2025.
6. Yadav V., Nath S., Shekhar, S., and Dikshit, O., Integrated landslide hazard zonation using geospatial techniques and DInSAR in the Rudraprayag District, Uttarakhand, 2025 IEEE India Geoscience and Remote Sensing Symposium (InGARSS 2025), 2025.
7. Nath S., Shekhar, S., and Dikshit, O., Subsidence detection in groundwater-stressed regions of Delhi–Haryana using Sentinel-1A DInSAR, 2025 IEEE India Geoscience and Remote Sensing Symposium (InGARSS 2025), 2025.
8. Nagarajan R., Nath, S., Sharma, A., Shekhar, S., Nagaraj, B., and Dikshit, O., Development of a

geometric local geoid model using GNSS/levelling observations, 2025 IEEE India Geoscience and Remote Sensing Symposium (InGARSS 2025), 2025.

9. Satyam Kumar, V., Nath, S., Shekhar, S., Sharma, S., and Dikshit, O., Calibration of UAVSAR data: Radiometric correction and Ainsworth-based polarimetric compensation, 2025 IEEE India Geoscience and Remote Sensing Symposium (InGARSS 2025), 2025.

10. P. Gandugade, S. Shekhar, S. Nath, B. Nagarajan and O. Dikshit, Advancing India's Geospatial Science: The Imperative for Satellite Laser Ranging (SLR) Infrastructure, 2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS), Goa, India, 2024, pp. 1-4, doi: 10.1109/InGARSS61818.2024.10984233.

11. S. V. Mishra, S. Shekhar, D. K. Pandey, R. Prakash, S. Nath and O. Dikshit, Machine Learning Driven Enhancement of Soil Moisture using CYGNSS GNSS-R Data and Meteorological Observations over Indian Agriculture Site, 2024 IEEE India Geoscience and Remote Sensing Symposium (InGARSS), Goa, India, 2024, pp. 1-4, doi: 10.1109/InGARSS61818.2024.10984217.

12. S. Nath, R. S. Chatterjee, O. Dikshit, S. Shekhar and S. P. Mohanty, DInSAR-Based Long-Term Deformation Monitoring in the NW Himalaya: Implications for Tectonic Activity, 2024 International Conference on Advances in Computing, Communication and Materials (ICACCM), Dehradun, India, 2024, pp. 1-5, doi: 10.1109/ICACCM61117.2024.11059136.

13. S. Nath, A. Sharma, S. Shekhar and O. Dikshit, Deformation analysis in the Central Himalaya inferred from InSAR technique: A case study of Nepal Earthquake, 2024 International Conference on Advances in Computing, Communication and Materials (ICACCM), Dehradun, India, 2024, pp. 1-5, doi: 10.1109/ICACCM61117.2024.11059160.

14. S. V. Mishra, S. Shekhar, D. K. Pandey, R. Prakash, S. Nath, and O. Dikshit, "Machine Learning Driven Enhancement of Soil Moisture Using CYGNSS GNSS-R Data and Meteorological Observations Over Indian Agriculture Site," in *Proc. IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*, 2024, pp. 1–4.

15. M. Kumar, A. Sharma, S. Nath, and S. Shekhar, "Tectonic Plate Interactions and Seismic Hazard Prediction: A Detailed Study Using GPS Stress Vectors and Geophysical Data," in *Proc. IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*, 2024, pp. 1–4.

16. Y. Ketholia, A. Joshi, S. Shekhar, and S. Nath, "Deciphering Earthquake Precursors: Unraveling the Nexus of Thermal Anomalies, Seismic Activity, and Satellite Data in Ladakh," in *Proc. IEEE India Geoscience and Remote Sensing Symposium (InGARSS)*, 2024, pp. 1–4.

17. K. S. Dhami, A. Debnath, and S. Shekhar, "Seismic Analysis of a G+9 Multi-Story Building: Comparative Study Using Linear Static and Response Spectrum Methods," in *Proc. Int. Conf. Advances in Computing, Communication and …*, 2024.

18. S. Nath, R. S. Chatterjee, O. Dikshit, S. Shekhar, and S. P. Mohanty, "DInSAR-Based Long-Term Deformation Monitoring in the NW Himalaya: Implications for Tectonic Activity," in *Proc. Int. Conf. Advances in Computing, Communication and …*, 2024.

19. S. Nath, A. Sharma, S. Shekhar, and O. Dikshit, "Deformation Analysis in the Central Himalaya Inferred from InSAR Technique: A Case Study of Nepal Earthquake," in *Proc. Int. Conf. Advances in Computing, Communication and …*, 2024.

20. S. Shekhar, R. Prakash, D. K. Pandey, A. Vidyarthi et al., " Vegetation-specific Correction for Improved Soil Moisture Estimation through Multipath Phase Analysis using NavIC-IC," 2024 IEEE International Geoscience and Remote Sensing Symposium IGARSS, 2024.

21. S. Rai, **S. Shekhar**, A. Vidyarthi, R. Prakash and R. Gowri, " Investigation of Machine Learning Approaches for Retrieval of Soil Moisture using NavIC Signal," 2023 IEEE International Conference On Electrical, Electronics, Communication and Computers (ELEXCOM), IIT ROORKEE, India, 2023.

22. **S. Shekhar**, R. Gusain, A. Vidhyarthi and R. Prakash, "Role of Remote Sensing and GIS Strategies to Increase Crop Yield," 2022 International Conference on Advances in Computing, Communication and Materials (ICACCM), Dehradun, India, 2022, pp. 1-6.

23. **S. Shekhar**, R. Prakash, D. K. Pandey, A. Vidyarthi et al., "Sensitivity of Multipath Peak Frequency of Navigation with Indian Constellation (NavIC) towards Surface Soil Moisture over Bare Land," 2021 IEEE International Geoscience and Remote Sensing Symposium IGARSS, 2021, pp. 7000-7003.
24. **S. Shekhar**, R. Prakash, D. K. Pandey, A. Vidyarthi, D. Putrevu and A. Misra, "Comparative Analysis of NavIC Multipath Amplitude and Phase for Soil Moisture Sensitivity over Different land cover," 2021 IEEE International India Geoscience and Remote Sensing Symposium (InGARSS), 2021, pp. 265-268.
25. **S. Shekhar**, R. Prakash, D. K. Pandey, A. Vidyarthi, D. Putrevu and A. Misra, "Sensitivity Analysis of GNSS-IR based Multipath Phase for Soil Moisture over Winter Wheat crop using Navigation with Indian Constellation (NavIC)," 2021 IEEE International India Geoscience and Remote Sensing Symposium (InGARSS), 2021, pp. 512-515.
26. **S. Shekhar**, R. Prakash, A. Vidyarthi and D. K. Pandey, "Sensitivity Analysis of Navigation with Indian Constellation (NavIC) Derived Multipath Phase Towards Surface Soil Moisture Over Agricultural Land," 2020 6th International Conference on Signal Processing and Communication (ICSC), 2020, pp. 138-142.
27. S. C. Bhardwaj, **S. Shekhar**, A. Vidyarthi and R. Prakash, "Satellite Navigation and Sources of Errors in Positioning: A Review," 2020 International Conference on Advances in Computing, Communication & Materials (ICACCM), Dehradun, India, 2020, pp. 43-50.